

Questions: Learning to Invent

Analytical Questions

1. Distinguish between parameter learning, structure learning, and policy learning in Active Inference. For each mode, provide a specific example of how it applies to the development of inventive skill.
2. How does deliberate practice differ from mere repetition in the context of invention? Use the Active Inference framework to explain why targeted exercises with feedback produce better inventive skill development than simply inventing more.
3. Analyze a patent as an Active Inference artifact. What do the "Background of the Invention," "Problem to be Solved," and "Detailed Description" sections correspond to in terms of generative model, prediction error, and model update?
4. Why does biomimicry emphasize functional principles over morphological copying? Use the Active Inference concepts of structural versus surface features to explain why copying a bird's wing shape often fails while copying the aerodynamic principle often succeeds.
5. The learning spiral (perception-cognition-action-learning) suggests that all four phases are necessary for inventive skill development. What goes wrong when an inventor skips the action phase (reads and thinks but does not build)? What goes wrong when they skip the cognition phase (builds without reasoning about why)?
6. Edison maintained detailed experimental notebooks and daily experiment quotas. Analyze this practice using the Active Inference framework. How did systematic record-keeping enable more effective model updating than ad hoc experimentation?
7. TRIZ was developed by analyzing over 200,000 patents to identify recurring solution patterns. Explain this as a form of structure learning at the meta-level — learning how to update models by studying examples of successful model updating.

8. The Eastgate Centre in Harare was designed by learning from termite mound thermodynamics. Identify the specific structural principles that transferred (what was learned from nature) and the aspects that required new engineering (what nature's model did not directly provide).
9. Compare the learning strategies of a novice inventor and an expert inventor using the concepts of parameter learning, structure learning, and policy learning. How does the balance between these three modes shift as expertise develops?
10. The concept of "kaizen" (continuous improvement) in Toyota's production system represents a systematic approach to inventive learning at the organizational level. Analyze how kaizen implements the perception-action-learning loop at scale. What makes it effective as a collective learning system?

Applied Questions

1. Conduct a self-assessment of your inventive skills across perception, cognition, action, and learning. Rate each competency area (1-10) and identify your three biggest weaknesses. What type of learning (parameter, structure, or policy) does each weakness require?
2. Search a patent database for one patent related to your invention domain. Read it through the Active Inference lens and extract: the prior model (what was assumed before), the prediction error (what failed), and the updated model (what the invention teaches). State the inventive principle as a general rule.
3. Describe the core functional challenge of your invention. Identify two biological organisms that face a structurally similar challenge. For the most promising analogy, describe how the biological principle could be translated to your engineering domain.
4. Design a deliberate practice exercise that targets your weakest inventive skill. Specify what you would do, how often, how you would get feedback, and how you would measure improvement over time.

5. Reflect on your recent inventive work. Which phase of the learning spiral do you spend the most time in, and which do you neglect? Design a specific intervention to activate the neglected phase this week.
6. Study a failed invention (one that was built but did not succeed in the market or in practice). Identify the generative model error that caused the failure. What should the inventor have learned from available evidence that they missed?
7. Create a "learning from nature" inventory for your invention domain. List five biological systems that are relevant to the functional challenges in your domain. For each, identify the structural principle and assess how directly it could transfer to engineering.
8. Design a weekly learning schedule that includes at least one activity from each phase of the learning spiral. Specify the duration, the materials needed, and the expected learning outcome for each activity.
9. Compare your learning strategy over the past month with the ideal learning spiral. How many full cycles (perception-cognition-action-learning) did you complete? What would you need to change to complete at least one full cycle per week?
10. Write a "learning journal entry" for your most recent inventive session. Document what you observed (perception), what you reasoned about (cognition), what you built or tested (action), and what you learned (model update). Then identify what the next session should focus on based on the remaining uncertainties.